

Lesson 6 PWM Servo Speed Control

This lesson focuses on controlling the No.1 PWM interface on the expansion board to control the speed of the servo by adjusting its angle. This adjustment relies on altering the time it takes for the servo to reach the desired angle position. For detailed principle instruction of PWM servo, please refer to “4.Expanded Lesson\3.Raspberry Pi Expansion Board Lesson\2.Expansion Board Control Lesson\Lesson 5 Single PWM Servo Control”.

1. Working Principle

In the program, modify the runtime, servo ID, and servo pulse width to control the rotation speed of the servo. The source code of the program is located in “/home/pi/board_demo/pwm_servo_speed.py”.

```
if __name__ == '__main__':
    while True:
        board.pwm_servo_set_position(0.5, [[1, 1500]]) # 设置1号舵机脉宽为1500(set the pulse width of servo 1 to 1500)
        time.sleep(0.5)
        board.pwm_servo_set_position(0.5, [[1, 1000]]) # 设置1号舵机脉宽为1000(set the pulse width of servo 1 to 1000)
        time.sleep(0.5)
        board.pwm_servo_set_position(0.5, [[1, 500]]) # 设置1号舵机脉宽为500(set the pulse width of servo 1 to 500)
        time.sleep(0.5)
        board.pwm_servo_set_position(1, [[1, 1000]]) # 设置1号舵机脉宽为1000(set the pulse width of servo 1 to 1000)
        time.sleep(1)
        board.pwm_servo_set_position(1, [[1, 1500]]) # 设置1号舵机脉宽为1500(set the pulse width of servo 1 to 1500)
        time.sleep(1)
        board.pwm_servo_set_position(1, [[1, 2000]]) # 设置1号舵机脉宽为2000(set the pulse width of servo 1 to 2000)
        time.sleep(1)
        board.pwm_servo_set_position(1, [[1, 2500]]) # 设置1号舵机脉宽为2500(set the pulse width of servo 1 to 2500)
        time.sleep(1)
        if not start:
            board.pwm_servo_set_position(1, [[1, 1500]]) # 设置1号舵机脉宽为1500(set the pulse width of servo 1 to 1500)
            time.sleep(1)
            print('已关闭(closed)')
            break
```

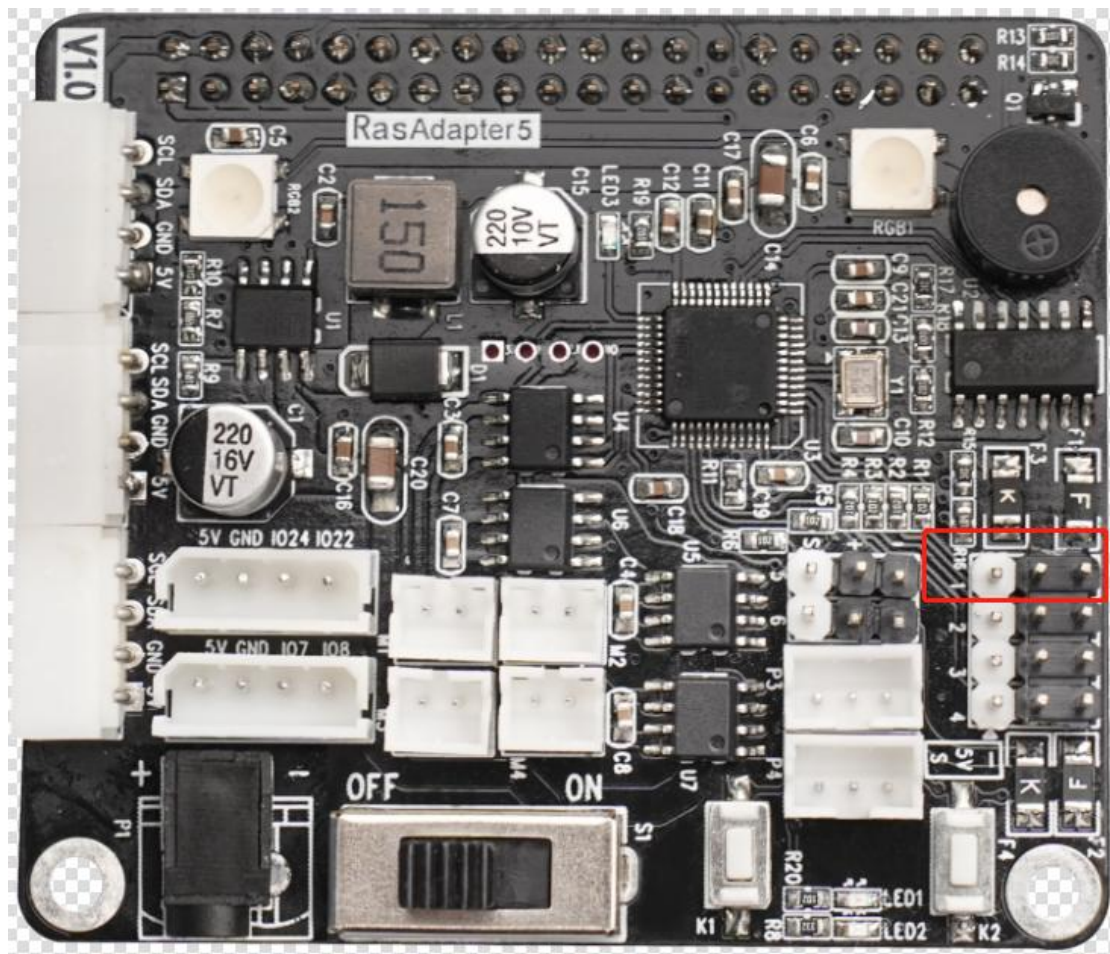
The program calls the “pwm_servo_set_position()” function in the “Board” library to implement the servo control. Take “board.pwm_servo_set_position(0.5, [[1, 1500]])” as an example, among them:

The first parameter “0.5” indicates the servo’s runtime is 500ms. If it is “1”, indicating the runtime is 1000ms.

The second parameter “[[1, 1500]]” represents setting the No.1 servo’s pulse width as 1500.

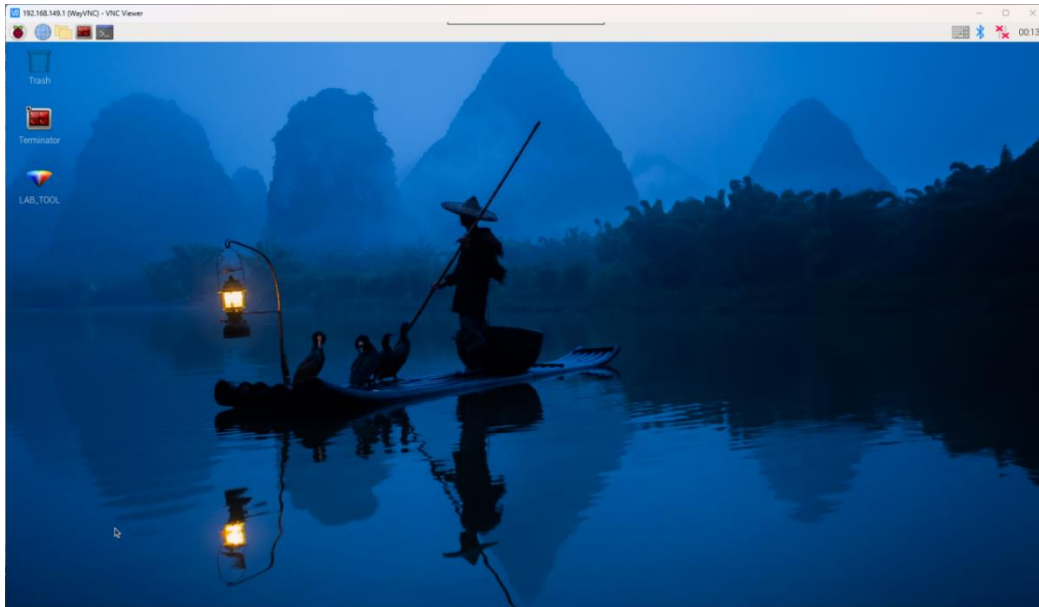
2. Getting Ready

There are 6 PWM servo interfaces on the Raspberry Pi expansion board. Let’s test by connecting the No.1 servo, as shown below:

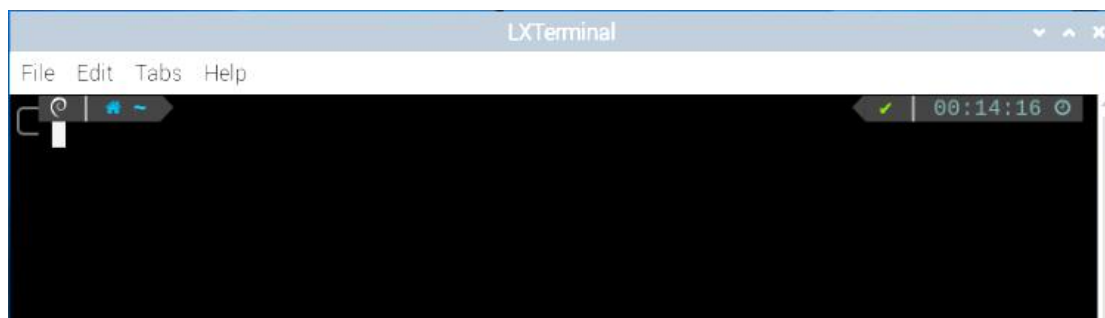


3. Operation Steps

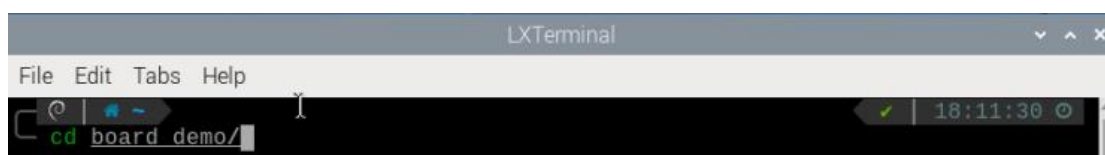
1) Turn the device on and connect the robot through the VNC remote connection tool.



2) Click  in the top left corner of the desktop, or press the shortcut key "Ctrl+Alt+T" to open the command line terminal.



3) Enter the command "cd board_demo/" to navigate to the directory where the demo program is located, then press "Enter".



Enter the command of running the program "python3 pwm_servo_speed.py", and press "Enter".

```
pi@raspberrypi:~/board_demo $ python3 pwm_servo_speed.py
*****
*****功能:幻尔科技树莓派扩展板, 控制PWM舵机速度(function:Hiwonder Raspberry Pi expansion board, PWM servo speed control)*****
*****
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Official website:https://www.hiwonder.com
Online mall:https://hiwonder.tmall.com
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Tips:
* 按下Ctrl+C可关闭此次程序运行, 若失败请多次尝试! (Press "Ctrl+C" to exit the program. If it fails, please try again multiple times!)
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```

4) To close the program, press “Ctrl+C”. If the program cannot be closed successfully, repeat this operation until it exits.

4. Program Outcome

Once the program is executed, the Raspberry Pi expansion board can adjust the pulse width of the servo to control the time it takes to reach a specific angle, This enables speed control of the servo motor.